

Gila Monster — Pulse generator v1.0

Bench Test Report

Unit under test: Gila Monster v1.0, single output port (D7), Arduino Uno R3 (ATmega328P, 16 MHz).

Instruments: digital oscilloscope, 1 M Ω DC-coupled probe on D7; firmware session report over the serial port (Comm, 9600 baud, 8-N-1).

Session length: 300 s (Timer = 300,000 ms) unless stated otherwise.

Reported quantities: *Pulses fired* and *Timing faults* are taken from the firmware session report. A timing fault is an edge placed more than 50 μ s after its scheduled time.

1. Edge timing

Quantity	Specification	Measured
Pulse width	programmed value $\pm 4 \mu$ s	within $\pm 4 \mu$ s
Period	programmed value $\pm 4 \mu$ s	within $\pm 4 \mu$ s
Dominant jitter location	—	falling edge

Edge placement resolution is bound by the 4 μ s step of the micros() time base. Rising edges are scheduled on an absolute time grid; falling edges are referenced to the rising edge and therefore do not always coincide with the grid.

2. Continuous mode

#	Configuration	Expected	Measured	Faults
1	500.00 Hz, Fixed 0.10 ms (T = 2,000 μ s, PW = 100 μ s)	150,000 pulses	150,000	0
2	0.10 Hz, Fixed 100.00 ms (T = 10 s, PW = 100 ms)	30 pulses	30	0
3	20.00 Hz, Fixed 5.00 ms (duty 10 %)	6,000 pulses	6,000	0
5	100.00 Hz, Duty Cycle 50.00 %	PW = 5,000 μ s	5,000 μ s, 30,000 pulse	0
17	20.00 Hz, Fixed 1.00 ms, Timer 600,000 ms	12,000 pulses	12,000	0

3. Parameter limits

#	Configuration	Expected	Measured
4	500.00 Hz, Fixed 5.00 ms requested	clamped to 1.90 ms ($T - 100 \mu s$), Auto-adjusted	clamped to 1.90 ms, Auto-adjusted shown
6	100.00 Hz, Duty Cycle 0.01 % requested	clamped to 1.00 % ($100 \mu s$ floor)	clamped to 1.00 %, HIGH time $100 \mu s$

In both cases the limit is applied to the displayed value, and the output matches the display.

4. Burst mode

#	Configuration	Expected	Measured	Faults
7	50.00 Hz, Count 10, Gap 1,818 ms, PW 2.00 ms	1,500 pulses (150 trains)	1,500	0
8	50.00 Hz, Count 1, Gap 0 ms	1 pulse, generator switches OFF	1	0
9	100.00 Hz, Count 999, Gap 5,000 ms, PW 2.00 ms	20,016 pulses	20,016	0

Test 7: $\text{cycle} = (10 - 1) \times 20 \text{ ms} + 2 \text{ ms} + 1,818 \text{ ms} = 2,000 \text{ ms}$; 150 complete cycles in 300 s. The train that would start exactly at the end of the session is not emitted.

Test 9: $\text{cycle} = (999 - 1) \times 10 \text{ ms} + 2 \text{ ms} + 5,000 \text{ ms} = 14,982 \text{ ms}$; 20 complete cycles (19,980 pulses) plus 36 pulses of the train cut by the end of the session.

5. NPS mode

#	Configuration	Expected	Measured	Faults
10	Count 4, ITImin 20 ms, PW 0.10 ms	1,200 pulses, no interval < 20 ms	1,200, constraint met	0
11	Count 50, ITImin 19 ms, PW 0.05 ms	15,000 pulses	15,000	0
12	Count 1, ITImin 999 ms, PW 0.05 ms	300 pulses, intervals of 1,000 ms	300, all ITI 1,000 ms	0

Test 11 is the upper bound of the mode: $50 \times 19 \text{ ms}$ occupies 95 % of the window, so most draws fall below the minimum interval and are pushed to it, and the train degenerates towards a periodic one.

6. Session Timer

#	Configuration	Expected	Measured	Faults
13	Cont. 1.00 Hz, Timer 10,000 ms	10 pulses (at 0, 1, ... 9 s)	10	0
14	Burst 2.00 Hz, Count 10, Gap 1,000 ms, PW 100.00 ms, Timer 3,300 ms	7 pulses, last pulse not truncated	7, full width	0
14b	Cont. 1.00 Hz, PW 500.00 ms, Timer 2,500 ms	3 pulses, third one complete	3, full width	0

Only pulses whose rising edge is scheduled before the end of the session are emitted. A pulse already in progress is always completed.

7. Trigger mode and manual firing

#	Configuration	Expected	Measured
15	Trigger, Cont. 1.00 Hz, Timer 10,000 ms	trigger starts the session; triggers during the session ignored; next trigger starts a new session	as expected
16	State OFF, Single-Shot button (D6)	one pulse per press, programmed width, no repeat while held	as expected

8. Summary

All 17 tests met their expected result. Pulse counts matched the computed values exactly in every mode, including the cases where the session Timer cuts a burst. No timing faults were reported in any session, including sessions of 150,000 pulses. Measured pulse width and period were within $\pm 4 \mu\text{s}$ of the programmed values, consistent with the resolution of the time base.

Not covered by this report: absolute clock accuracy (the Arduino Uno R3 ceramic resonator is typically within $\pm 0.5\%$, and all timings scale with it), which requires an external time reference; and the statistical verification of the NPS interval distribution, which is done offline from the Comm export.